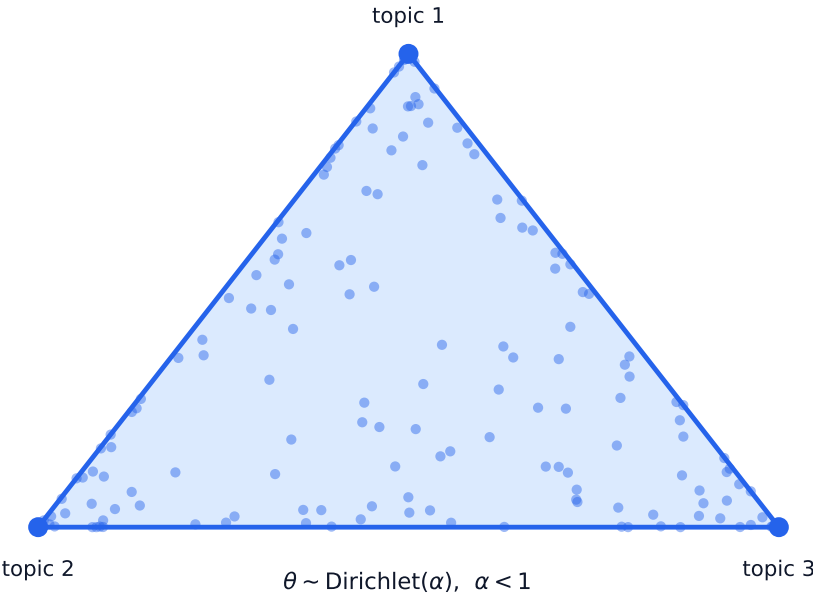
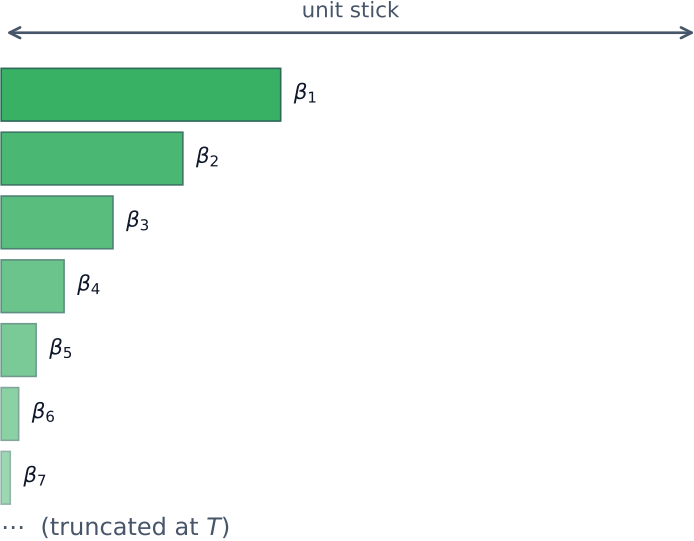


Four backbone priors impose four different geometries on θ
Same vocabulary, different generative assumption $\Rightarrow$ recipe re-ranking (the P4 mechanism)

LDA — Dirichlet simplex

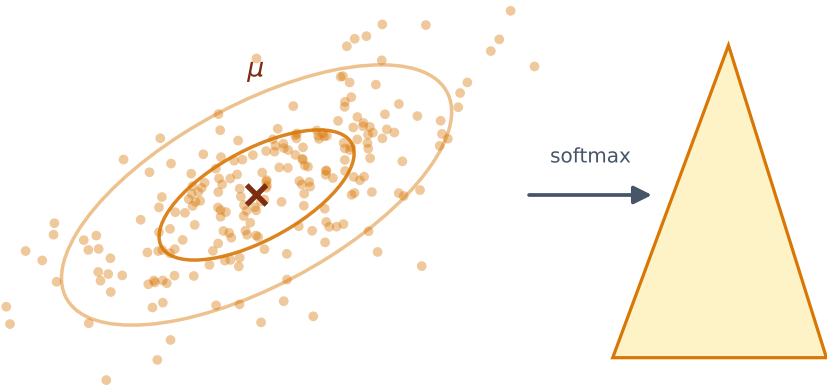


HDP — stick-breaking (GEM)



$$\beta_k = v_k \prod_{j < k} (1 - v_j), \quad v_k \sim \text{Beta}(1, \gamma)$$

ProdLDA — logistic-normal



$$\eta \sim \mathcal{N}(\mu, \Sigma), \quad \theta = \text{softmax}(\eta)$$

ETM — embedding directions

